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LAB ASSIGNMENT – 09

Consider the following tables namely "DEPARTMENTS" and "EMPLOYEES" Their schemas are as follows,

Departments (dept no , dept name , dept_location); Employees (emp_id , emp_name , emp_salary,dept_no);

```
CREATE DATABASE KHUSHI_LABASSIGN_9;
USE KHUSHI_LABASSIGN_9;
CREATE TABLE DEPARTMENTS (
    dept_no INT PRIMARY KEY,
    dept_name VARCHAR(50),
    dept_location VARCHAR(50)
);
CREATE TABLE EMPLOYEES (
    emp_id INT PRIMARY KEY,
    emp_name VARCHAR(50),
    emp_salary DECIMAL(10,2),
    dept_no INT,
    FOREIGN KEY (dept_no) REFERENCES DEPARTMENTS(dept_no)
);
```

(a) Develop a query to grant all privileges of employees table into departments table.

FIRST WE HAVE TO CREATE A USER.

```
CREATE USER 'departments_user'@'localhost'
IDENTIFIED BY '1234';
```

THEN WE GRANT ALL PRIVILEGES ON THE EMPLOYEE TABLE TO DEPARTMENTS .

```
GRANT ALL PRIVILEGES
ON KHUSHI_LABASSIGN_9.EMPLOYEES
TO 'departments_user'@'localhost';
SHOW GRANTS FOR 'departments_user'@'localhost';
```

OUTPUT:

Result Grid | Filter Rows:

Grants for departments_user@localhost
GRANT USAGE ON *.* TO `departments_user`...
GRANT ALL PRIVILEGES ON `khushi_labassign_...

(b) Develop a query to grant some privileges of employees table into departments table.

```
GRANT SELECT, INSERT, UPDATE
ON KHUSHI_LABASSIGN_9.EMPLOYEES
TO 'departments_user'@'localhost';
SHOW GRANTS FOR 'departments_user'@'localhost';
```

OUTPUT:

Result Grid | Filter Rows:

Grants for departments_user@localhost
GRANT USAGE ON *.* TO `departments_user`...
GRANT ALL PRIVILEGES ON `khushi_labassign_...

(c) Develop a query to revoke all privileges of employees table from departments table.

```
REVOKE ALL PRIVILEGES
ON KHUSHI_LABASSIGN_9.EMPLOYEES
FROM 'departments_user'@'localhost';
SHOW GRANTS FOR 'departments_user'@'localhost';
```

OUTPUT:

Result Grid | Filter Rows:

Grants for departments_user@localhost
GRANT USAGE ON *.* TO `departments_user`...

(d) Develop a query to revoke some privileges of employees table from departments table.

First, we have to grant selected privileges on the EMPLOYEES table to DEPARTMENTS.

```
GRANT SELECT, INSERT, UPDATE
ON KHUSHI_LABASSIGN_9.EMPLOYEES
TO 'departments_user'@'localhost';
SHOW GRANTS FOR 'departments_user'@'localhost';
```

OUTPUT:

Result Grid | Filter Rows:

Grants for departments_user@localhost
GRANT USAGE ON *.* TO `departments_user`...
GRANT SELECT, INSERT, UPDATE ON `khushi_l...

Now, we revoke selected privileges on the EMPLOYEES table from DEPARTMENTS.

```
REVOKE SELECT, INSERT
ON KHUSHI_LABASSIGN_9.EMPLOYEES
FROM 'departments_user'@'localhost';
SHOW GRANTS FOR 'departments_user'@'localhost';
```

OUTPUT :

Result Grid | Filter Rows:

Grants for departments_user@localhost
GRANT USAGE ON *.* TO `departments_user`...
GRANT UPDATE ON `khushi_labassign_3`.`em...

(e) Write a query to implement the save point.

Step 1: Start Transaction

```
START TRANSACTION;
```

Step 2: Insert into DEPARTMENTS

```
INSERT INTO DEPARTMENTS VALUES (101, 'HR', 'Delhi');
INSERT INTO DEPARTMENTS VALUES (102, 'IT', 'Mumbai');
SELECT * FROM DEPARTMENTS;
```

OUTPUT:

Result Grid | Filter Rows:

dept_no	dept_name	dept_location
101	HR	Delhi
102	IT	Mumbai
NULL	NULL	NULL

Step 3: Insert into EMPLOYEES

```
INSERT INTO EMPLOYEES VALUES (1, 'Amit', 50000, 101);
INSERT INTO EMPLOYEES VALUES (2, 'Riya', 60000, 102);
SELECT * FROM EMPLOYEES;
```

OUTPUT:

Result Grid | Filter Rows:

emp_id	emp_name	emp_salary	dept_no
1	Amit	50000.00	101
2	Riya	60000.00	102
NULL	NULL	NULL	NULL

Step 4: First Update (Dept 101).

```
UPDATE EMPLOYEES
SET emp_salary = emp_salary + 10000
WHERE dept_no = 101;
SELECT * FROM EMPLOYEES;
SAVEPOINT sp1;
```

OUTPUT:

Result Grid | Filter Rows:

emp_id	emp_name	emp_salary	dept_no
1	Amit	60000.00	101
2	Riya	60000.00	102
NULL	NULL	NULL	NULL

Step 5: Second Update (Dept 102)

```
UPDATE EMPLOYEES
SET emp_salary = emp_salary + 5000
WHERE dept_no = 102;
SELECT * FROM EMPLOYEES;
```

OUTPUT:

Result Grid | Filter Rows:

emp_id	emp_name	emp_salary	dept_no
1	Amit	60000.00	101
2	Riya	65000.00	102
NULL	NULL	NULL	NULL

Step 11: Rollback to SAVEPOINT AND COMMIT

```
ROLLBACK TO sp1;
COMMIT;
```

OUTPUT :

Result Grid | Filter Rows:

emp_id	emp_name	emp_salary	dept_no
1	Amit	60000.00	101
2	Riya	60000.00	102
NULL	NULL	NULL	NULL